

# FoodLens

## AI-Powered Ingredient Scanner and Health Risk Scoring System

*Literature Survey — Capstone Project*

### 1. Introduction

This literature survey reviews existing research and systems relevant to the six core areas underlying FoodLens: optical character recognition (OCR) for food packaging, personalized nutrition and diet recommendation systems, front-of-pack nutrition labeling and scoring methodologies, crowdsourced food product databases, commercial food-scoring applications and their documented limitations, and the reliability of large language models (LLMs) in health-related information generation. Each section summarizes relevant work and identifies the specific gap that FoodLens is designed to address.

### 2. OCR-Based Ingredient Extraction from Food Labels

Nagayi et al. (2025) evaluated four open-source OCR engines — Tesseract, EasyOCR, PaddleOCR, and TrOCR — on 231 real-world food packaging images collected in South African retail environments, covering 11 languages. Their study found that Tesseract achieved the lowest character error rate among the engines tested, but all systems suffered reduced accuracy on curved packaging surfaces and multilingual labels, highlighting that real-world label conditions remain a significant challenge for OCR pipelines rather than a solved problem.

Assiri et al. (2025), published in the *Journal of Imaging*, explored using large language models to extract structured nutritional information directly from bilingual food label images, aiming to handle cases where labels mix two languages on the same package — a scenario highly relevant to Indian packaging, which often mixes English with regional scripts.

Guru et al. (2024), in *Frontiers in Health Informatics*, presented an AI-powered system combining OCR and NLP specifically to analyze packaged food ingredients and assess their health impact, allowing a user to scan packaging and instantly receive a nutritional review — a direct conceptual precursor to FoodLens's own scan-to-score pipeline, though their published work does not address personalization to individual health conditions.

#### Relevance to FoodLens

These studies confirm that OCR accuracy on real-world, multilingual, and non-uniform packaging remains an open technical challenge rather than a solved one, which validates FoodLens's approach of pairing OCR with an explicit confidence score shown to the user, and an NLP-based alias-resolution layer rather than assuming raw OCR output is immediately reliable.

### 3. Personalized Nutrition and Diet Recommendation Systems

A food recommender system targeted at patients with diabetes and hypertension used expert interviews with medical professionals to validate a personalized food advisory system, concluding that chronic-disease patients specifically need nutrient-based guidance tailored to their condition rather than generic dietary advice.

Thianniwet, Keawlamoon, and Phosaard (2025) proposed a context-aware personalized food recommendation framework for Type 2 diabetes management, evaluated across 25 use cases with a reported 100% acceptability rate among test scenarios, demonstrating that condition-specific personalization is both technically feasible and well-received by intended users.

A diabetes-focused food recommender system (DFRS) published in *PLOS Digital Health* used graph neural networks to generate personalized dietary recommendations, reporting measurable clinical improvements including reduced glycemic

variability, illustrating that personalization in this domain can extend to measurable health outcomes and is not merely a cosmetic feature.

#### **Relevance to FoodLens**

This body of work supports personalization as a well-established and clinically meaningful direction in food-related health technology. However, these systems are largely research prototypes focused on meal/diet recommendation for a single condition (usually diabetes), rather than point-of-purchase packaged product scoring personalized across multiple simultaneous conditions and allergies — the specific gap FoodLens addresses through its condition-multiplier and allergen-override design.

## **4. Front-of-Pack Nutrition Labeling and Scoring Systems**

Multiple large-scale European studies (conducted across France, Italy, Portugal, and a 12-country comparison) have evaluated the effectiveness of front-of-pack labels such as Nutri-Score, Multiple Traffic Lights, and Reference Intakes. Across these studies, Nutri-Score consistently outperformed alternative label formats in helping consumers correctly rank products by nutritional quality, and was associated with improved purchasing intentions, including among individuals with existing cardiometabolic disease.

However, a systematic review of Nutri-Score's real-world effectiveness found that while experimental (lab-based) studies show a positive effect on consumers' ability to judge nutritional quality, real-life studies show little to no measurable effect on actual eating patterns or health outcomes — indicating a meaningful gap between label comprehension and behavior change.

#### **Relevance to FoodLens**

This research validates the general principle that a simple, visual, color-coded score can meaningfully improve a consumer's ability to judge a product's healthiness at the point of decision — the same underlying principle behind FoodLens's traffic-light Health Risk Score. At the same time, the finding that label comprehension alone does not guarantee behavior change motivates FoodLens's additional consumption-tracking and reporting features, which aim to reinforce healthier choices over time rather than relying on a single point-of-purchase signal.

## **5. Crowdsourced Food Product Databases**

Open Food Facts, launched in 2012, is a free, collaborative, crowdsourced database of food products maintained under an open license, and has become a widely cited resource in nutrition and food-science research. Since 2019, the project has incorporated machine learning (via its Robotoff tool) to assist with image recognition and data extraction, supplementing manual crowdsourced contributions.

Independent comparisons of nutrition datasets note that despite its scale, Open Food Facts exhibits a packaged-food bias, with fresh produce, bulk foods, and regionally specific or small-manufacturer products significantly underrepresented, along with variable completeness across entries — directly corroborating the regional coverage gap FoodLens is designed to address for the Indian market.

#### **Relevance to FoodLens**

These findings both validate Open Food Facts as a suitable primary data source for FoodLens's barcode-lookup module and confirm the specific, documented gap (regional/local product underrepresentation) that FoodLens's community-sourced submission and consensus-verification feature is designed to close.

## 6. Commercial Food-Scoring Applications and Their Limitations

Yuka, a widely used commercial food and cosmetics scanning application, computes its score from a fixed weighting (nutritional quality, presence of additives, and organic status). An academic review of ethics in food recommender applications observed that, unlike Nutri-Score — which underwent nearly a decade of expert deliberation and formal validation by French regulatory bodies before adoption — Yuka's scoring methodology has not been subjected to comparable external scientific scrutiny.

This concern is echoed in professional nutrition commentary, which has noted that Yuka's additive risk classifications sometimes extend conclusions from limited or contested studies more broadly than the underlying evidence supports, and that its scoring does not account for portion size, overall dietary context, or individual health conditions.

### Relevance to FoodLens

This is a central motivating gap for FoodLens's own design: rather than relying on an opaque, internally weighted scoring formula, FoodLens ties every ingredient risk score to a specific, citable source (FSSAI/WHO/ICMR guidance) and documents its scoring methodology explicitly, while also directly addressing the personalization gap noted across this literature by adjusting scores to the individual user's health profile rather than applying one static formula to every user.

## 7. LLM Reliability and Grounding in Health-Related Applications

Recent research consistently identifies hallucination — the generation of plausible but factually incorrect content — as a central risk of applying large language models to health-related information tasks. Retrieval-augmented generation (RAG), which grounds LLM output in an explicitly retrieved, verified corpus rather than the model's internal parameters alone, has emerged as the dominant mitigation strategy across this literature.

Lee et al. (2024), published in the *Journal of Personalized Medicine*, developed a dual RAG system grounded in official diabetes management guidelines specifically to reduce hallucination in diabetes-related LLM outputs, demonstrating that domain-specific grounding measurably improves reliability over ungrounded LLM responses in a health context directly relevant to FoodLens's target users.

### Relevance to FoodLens

This research directly supports FoodLens's core architectural decision to constrain LLM usage strictly to phrasing already-verified, rule-based scoring output, rather than allowing the LLM to generate health claims independently — applying the same grounding principle this literature identifies as essential for reliable health-related AI output, implemented here through a deterministic scoring engine acting as the 'retrieved, verified' source instead of a document corpus.

## 8. Comparative Summary

| Existing Work / System                                                                                   | Approach & Limitation                                                                                                                                                             | How FoodLens Addresses It                                                                                                              |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| OCR studies on food packaging (Nagayi et al. 2025; Assiri et al. 2025)                                   | Demonstrate OCR accuracy drops on real-world, multilingual, curved-surface labels; no system specifically targets Indian regional packaging.                                      | OCR confidence score shown to user; NLP alias-resolution layer; community submission covers regions with poor label/database coverage. |
| Diabetes/condition-specific recommender systems (Thianniwet et al. 2025; DFRS, PLOS Digital Health 2025) | Effective personalization shown for single conditions in meal/diet recommendation; not applied to packaged-product point-of-purchase scoring or multiple simultaneous conditions. | Condition-multiplier system supports multiple simultaneous conditions and allergies at the point of scanning a packaged product.       |
| Nutri-Score and other front-of-pack labels (multi-country studies)                                       | Improves nutritional judgment in lab settings; limited evidence of real-world behavior change; identical score shown to every consumer.                                           | Same visual traffic-light principle, personalized per user, paired with consumption tracking to reinforce behavior over time.          |
| Open Food Facts                                                                                          | Large, open, crowdsourced dataset; documented packaged-food bias and regional/local product underrepresentation.                                                                  | Used as primary data source; community submission and consensus approval specifically targets underrepresented regional products.      |
| Yuka (commercial app)                                                                                    | Popular and usable, but scoring methodology not independently validated; additive risk classifications criticized as overstated; no personalization to health conditions.         | Scoring tied to explicit, citable regulatory/health sources (FSSAI/WHO/ICMR); fully personalized to user health profile.               |
| LLM/RAG reliability research (Lee et al. 2024, and related hallucination-mitigation studies)             | LLMs hallucinate health information when ungrounded; RAG grounding in verified sources measurably improves reliability.                                                           | LLM usage constrained strictly to phrasing pre-verified, rule-based scoring output — never generating health claims independently.     |

## 9. Research Gap and Conclusion

Across all six areas surveyed, no single existing system combines: (a) OCR/NLP-based extraction targeted at regionally underrepresented packaged products, (b) a transparent, source-cited rule-based scoring methodology rather than an opaque proprietary formula, (c) personalization across multiple simultaneous health conditions and allergies at the point of product scanning, and (d) LLM usage deliberately constrained to explanation-phrasing rather than independent health-claim generation. FoodLens is positioned specifically at the intersection of these four gaps, drawing on validated approaches from each body of literature while directly addressing the limitations documented within each.

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